Team 18



Crazy Dave Studios



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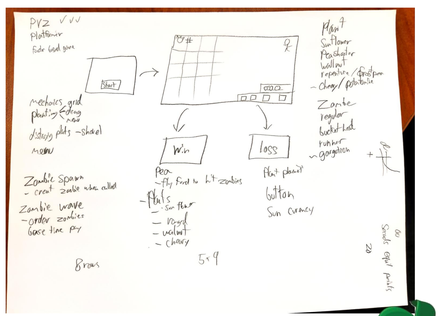
Operation Catapult 2026

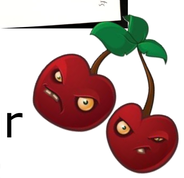
Design Process



The goal of our group was to make a fully playable remake of plants vs zombies. We did this by first brainstorming on which plants and zombies to implement. Then we had to decide how we would implement game mechanics and determine what we could reasonably accomplish during Operation Catapult.







We divided our process into 8 modules to allow us to work on many different part of the project and keep our code organized.

Key Features:

- plant placing (drag, drop, snap)
- zombie exponential spawning
- projectile-zombie interactions
- Zombie-plant interactions
- Currency system
- Start/death detections

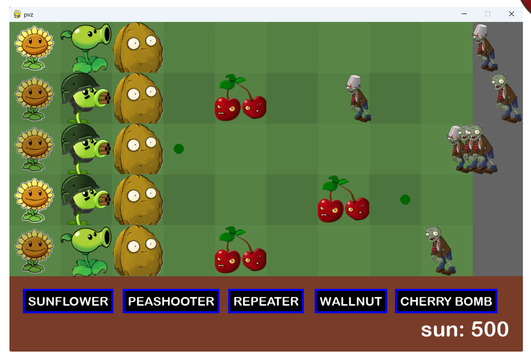






Final Product







```python
def main():
    poop = True
    fart = bool
    number = 0
    numex = 1
    total_spawns = 0
    pygame.init()

    try:
        pygame.mixer.init()
    except Exception:
        pass

music_available = pygame.mixer.get_init() is not None
music_tracks = ["assets/S-loonboon.mp3", "assets/hype_zombie_music.mp3"]
music_index = 0
MUSIC_END = pygame.USEREVENT + 1
music_started = False


pygame.display.set_caption("pvz")
screen = pygame.display.set_mode((1000, 650))

# load plant preview images
try:
    sunflower_preview = pygame.image.load("assets/sunflower.png").convert_alpha()
except Exception:
    sunflower_preview = None
selected_plant_image = None

wave = Zombie_wave_module.Zombie_wave(screen)

sun_button = Button_module.Button(screen, 115, 550, "sunflower")
pea_button = Button_module.Button(screen, 315, 550, "peashooter")
rep_button = Button_module.Button(screen, 500, 550, "repeater")
wall_button = Button_module.Button(screen, 665, 550, "wallnut")
cherry_button = Button_module.Button(screen, 850, 550, "cherry bomb")

start_button = Button_module.Button(screen, 500, 300, "play")
start_button.border_color = "white"
end_button = Button_module.Button(screen, 500, 300, "restart?")
end_button.border_color = "white"

first_screen = start_screen_module.start_screen(screen)
end_screen = start_screen_module.start_screen(screen)

sun_counter = money_module.money(screen)

clock = pygame.time.Clock()
all_plants = []
def plant_exists(row, col):
    return any(p.row == row and p.column == col for p in all_plants)

plant_cursor = plant_cursor_module.PlantCursor(screen)

while True:

    clock.tick(60)
    sun_amount = sun_counter.check_sun()

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            sys.exit()
        # handle end of music track to play next sequentially
        if event.type == pygame.USEREVENT + 1:
            try:
                music_index = (music_index + 1) % len(music_tracks)
                pygame.mixer.music.load(music_tracks[music_index])
                pygame.mixer.music.play()
```







Credit to George Fan and the team at PopCap Games for the inspiration for our game, the zombie and plant designs, and the game music.